RESEARCH METHODOLOGY IN PSYCHOLOGY

BY: AQSA FAYYAZ
Lecturer
(Science and Humanities)
FAST - National University of Computer & Emerging Sciences

LEARNING OUTCOMES

- What are scientific process?
- Types of researches
- Variables
- Research Methods
- Ethical Considerations in Research & applications of research

WHAT IS RESEARCH?

- **Systematic Inquiry:** Research follows a structured approach to collect and analyze information.
- **Knowledge Expansion:** It seeks to generate new insights, confirm existing theories, or explore different areas.
- **Problem Solving:** Research can address specific questions or issues, leading to practical solutions or advancements in a field.

TYPES OF RESEARCH

- **Basic Research:** Also known as fundamental or pure research, it focuses on expanding theoretical knowledge without immediate practical application. For example, studying cognitive processes to understand how memory works.
- **Applied Research:** It aims to solve specific, practical problems and often involves direct application of theories. For instance, developing new educational strategies based on cognitive research.

SCIENTIFIC PROCESS

1. Observation

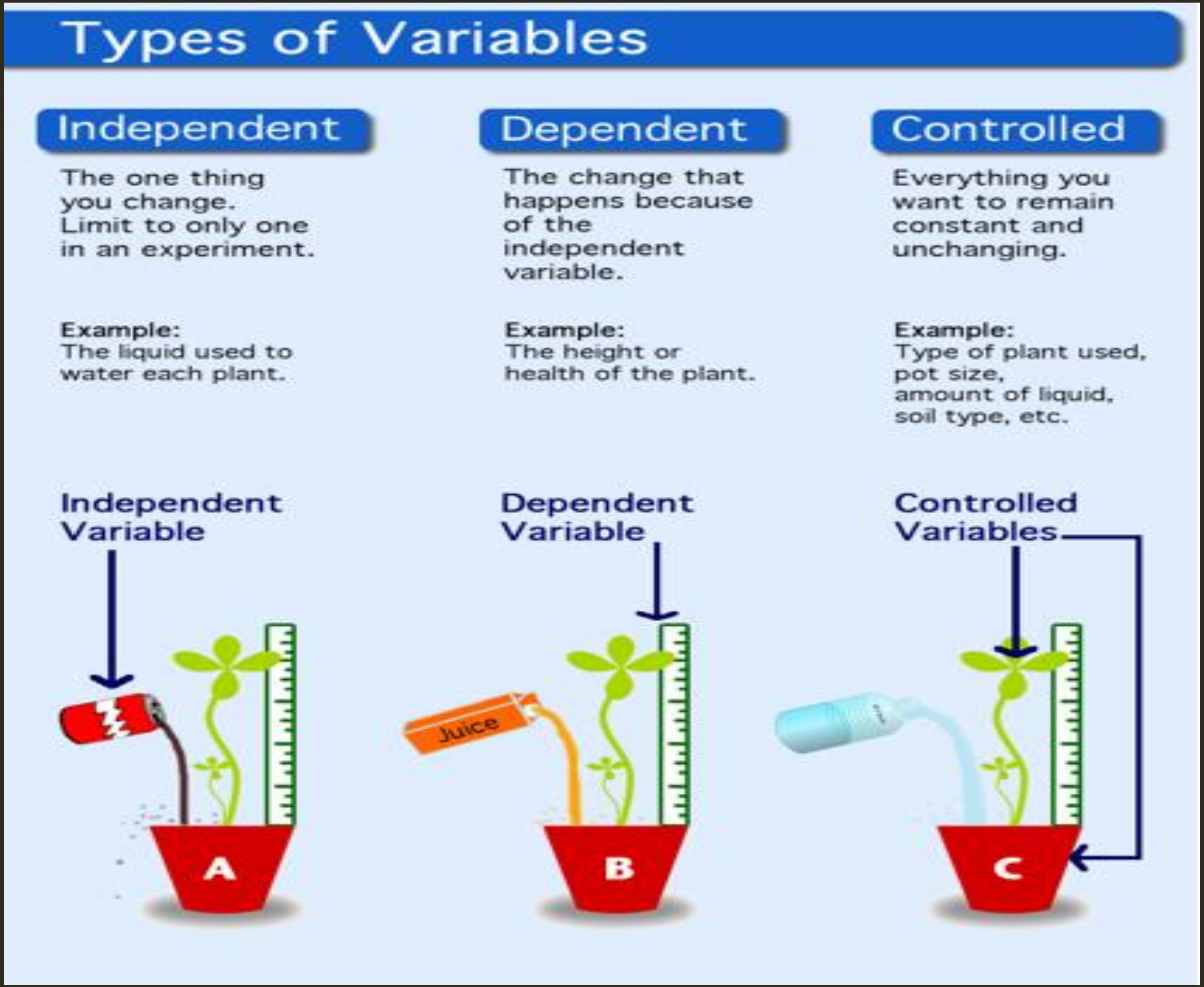
2. Formulating a Hypothesis: Proposing a tentative explanation or prediction based on the observations. A hypothesis should be specific, testable, and falsifiable.

3. Designing an Experiment:

Planning how to test the hypothesis, including selecting variables, controls, and methods for data collection.

•Variables:

- **Independent Variable:** The variable that is manipulated or changed in the experiment.
- **Dependent Variable:** The variable that is measured or observed to assess the effect of the independent variable.
- **Control Variables:** Factors that are kept constant to ensure that any changes in the dependent variable are due to the manipulation of the independent variable.


Types of Variables
Independent
The one thing you change. Limit to only one in an experiment.
Example: The liquid used to water each plant.
Dependent
The change that happens because of the independent variable.
Example: The height or health of the plant.
Controlled
Everything you want to remain constant and unchanging.
Example: Type of plant used, pot size, amount of liquid, soil type, etc.
Independent Variable
Dependent Variable
Controlled Variables
Juice
A
B
C

4. Conducting the Experiment

Execution: Performing the experiment according to the design, ensuring accurate and precise measurements and controls.

Data Collection: Systematically recording the results of the experiment.

5. Analyzing Data

Data Analysis: Using statistical methods or qualitative analysis to interpret the results and determine if they support or refute the hypothesis.

Comparison: Assessing the data against the predictions made by the hypothesis.

6. Drawing Conclusions

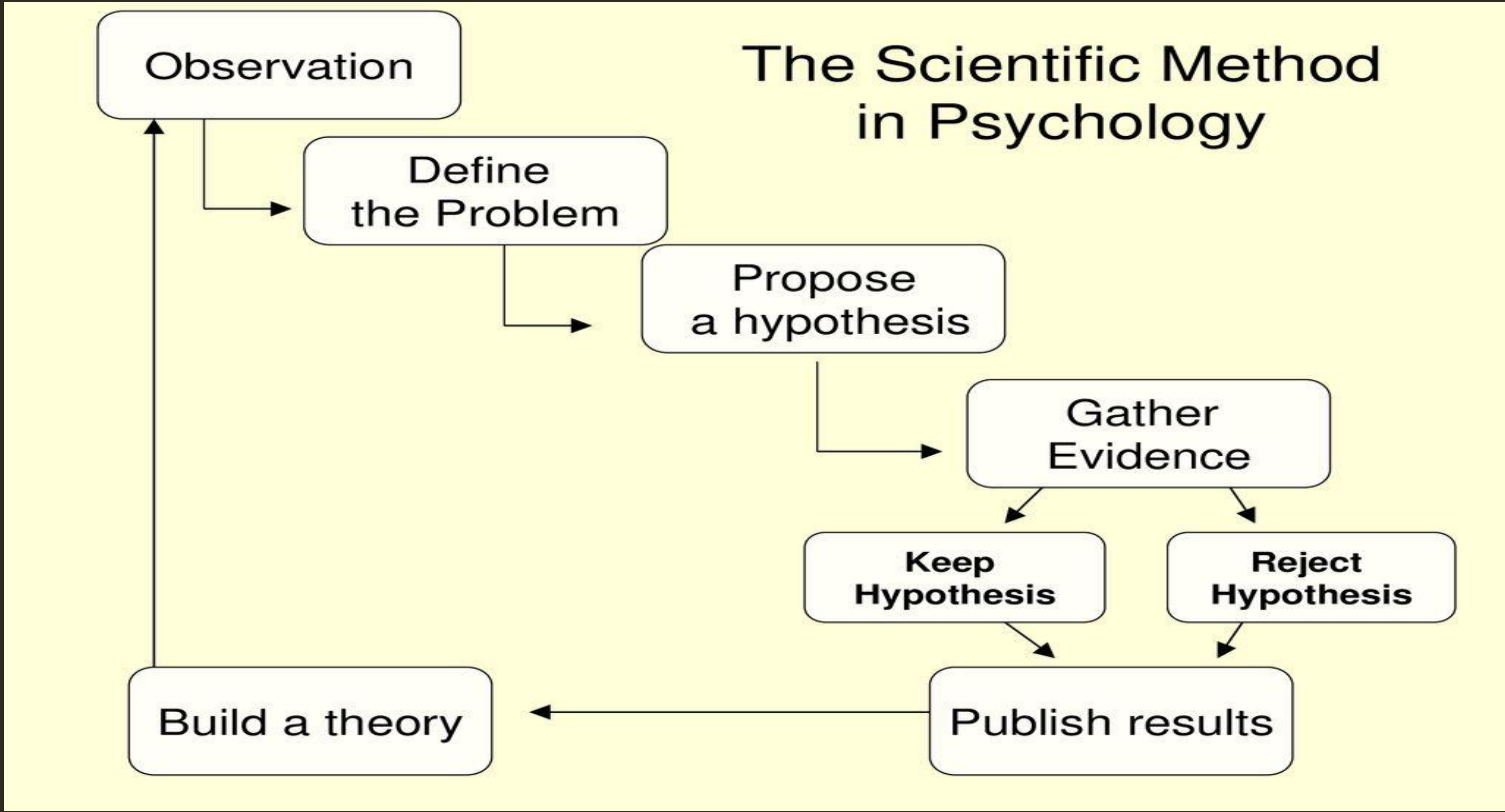
Conclusion Formulation: Based on the data analysis, determining whether the hypothesis is supported or rejected.

Implications: Discussing the significance of the findings, potential impacts, and how they relate to existing knowledge.

7. Reporting Results

Documentation: Writing detailed reports or papers that include the methodology, results, and conclusions. This often includes sharing the research in scientific journals or presentations.

Peer Review: Submitting the work for review by other experts in the field to validate the findings and methodology.


The Scientific Method
in Psychology
Observation
Define
the Problem
Propose
a hypothesis
Gather
Evidence
Keep
Hypothesis
Reject
Hypothesis
Publish results
Build a theory

RESEARCH PROCESS

1. **Identifying a Problem or Question:** Defining a clear and focused research question or problem.
2. **Conducting a Literature Review:** Reviewing existing research to understand what has already been discovered and to refine the research question.
3. **Designing the Study:** Developing a plan that includes choosing a research design, methods, and tools for data collection.
4. **Collecting Data:** Gathering information through various methods such as experiments, surveys, interviews, or observations.
5. **Analyzing Data:** Using statistical or qualitative methods to interpret the data and draw conclusions.
6. **Reporting Results:** Presenting findings in a structured format, often including a discussion of implications and recommendations.

RESEARCH METHODOLOGIES

- **Qualitative Research:** Focuses on understanding phenomena from a subjective perspective, often using methods like interviews, thematic analysis and case study researches
- **Quantitative Research:** Involves measuring and analyzing numerical data to identify patterns, relationships, or causal effects, using statistical techniques.
- **Mixed Methods Research:** Combines both qualitative and quantitative approaches to provide a comprehensive analysis.

ETHICAL CONSIDERATIONS

- **Informed Consent:** Ensuring participants are aware of the research and agree to participate voluntarily.
- **Confidentiality:** Protecting the privacy of participants and their data.
- **Integrity:** Conducting research honestly and accurately, avoiding fabrication or falsification of data.

APPLICATIONS OF RESEARCH

- **Academic and Scientific:** Contributing to the body of knowledge in various fields such as psychology, medicine, or sociology.
- **Professional and Practical:** Informing practices and policies in areas like education, healthcare, or business.